Holoqubed Atlas Engine

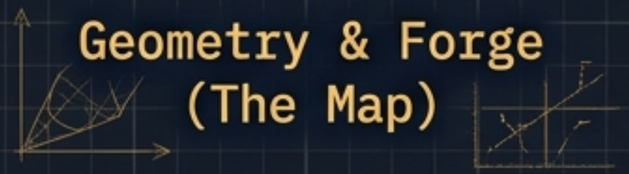
Geometric MLP routing for classical inference, and a real-amplitude quantum weight format emulated in PyQrack.

System Telemetry / Baseline: Every figure below was measured on a 2 vCPU VM (OPT-125m, OPT-350m, Qwen3-0.6B, WikiText-2).

Gemini Notebook

Architecture Framework: Geometry, Execution, Interfaces

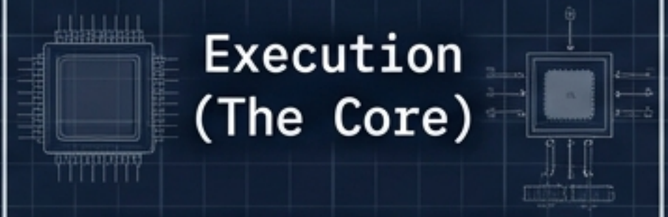
Geometry & Forge (The Map)



- Low-rank chart: PCA subspace of MLP inputs (rank 128)
- 64 Voronoi regions per layer (k-means on chart)
- Residual ellipsoid spread per region/neuron

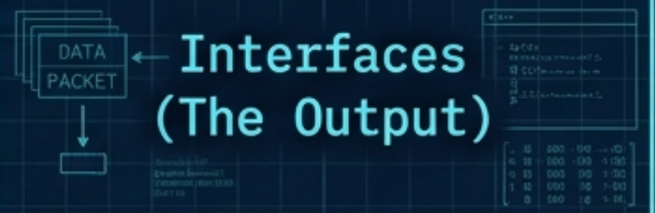


Execution (The Core)



- Atlas backends: NumPy reference & OpenCL kernels (multi-device round robin)
- Dense torch backend (for comparison)
- PyQrack: QrackSimulator tiles, QrackNeuron gating, QrackBridge patch chains

Interfaces (The Output)

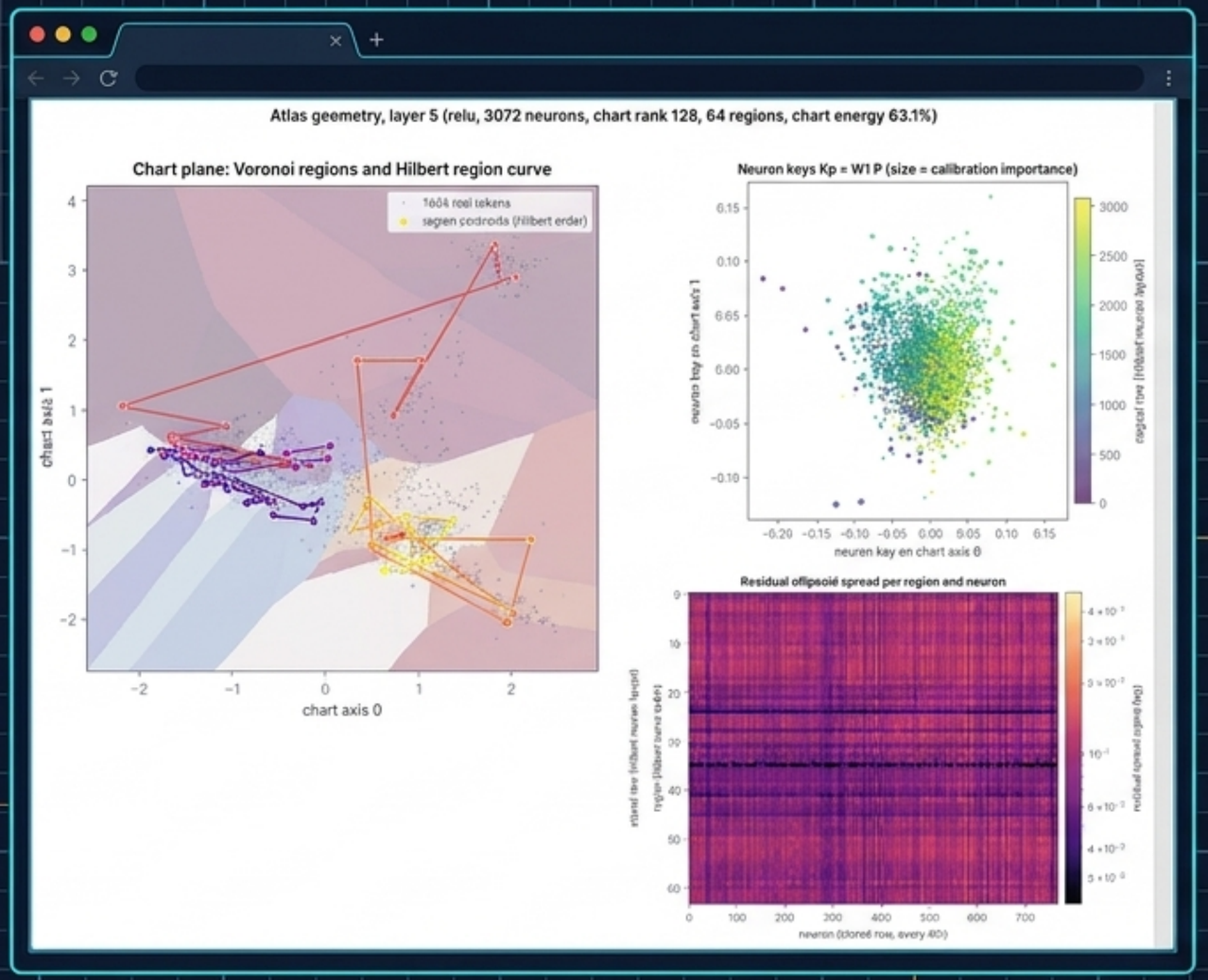


- OpenAI-compatible API with SSE streaming
- Live dashboard (browser, realtime)
- Landscape, qneuron, qconvert, and precision reports



Dependencies: No CUDA. No C++ build steps. Runs on OpenCL (GPU or PoCL on CPU) via torch, transformers, pyopencl, pyqrack.

Gemini Notebook

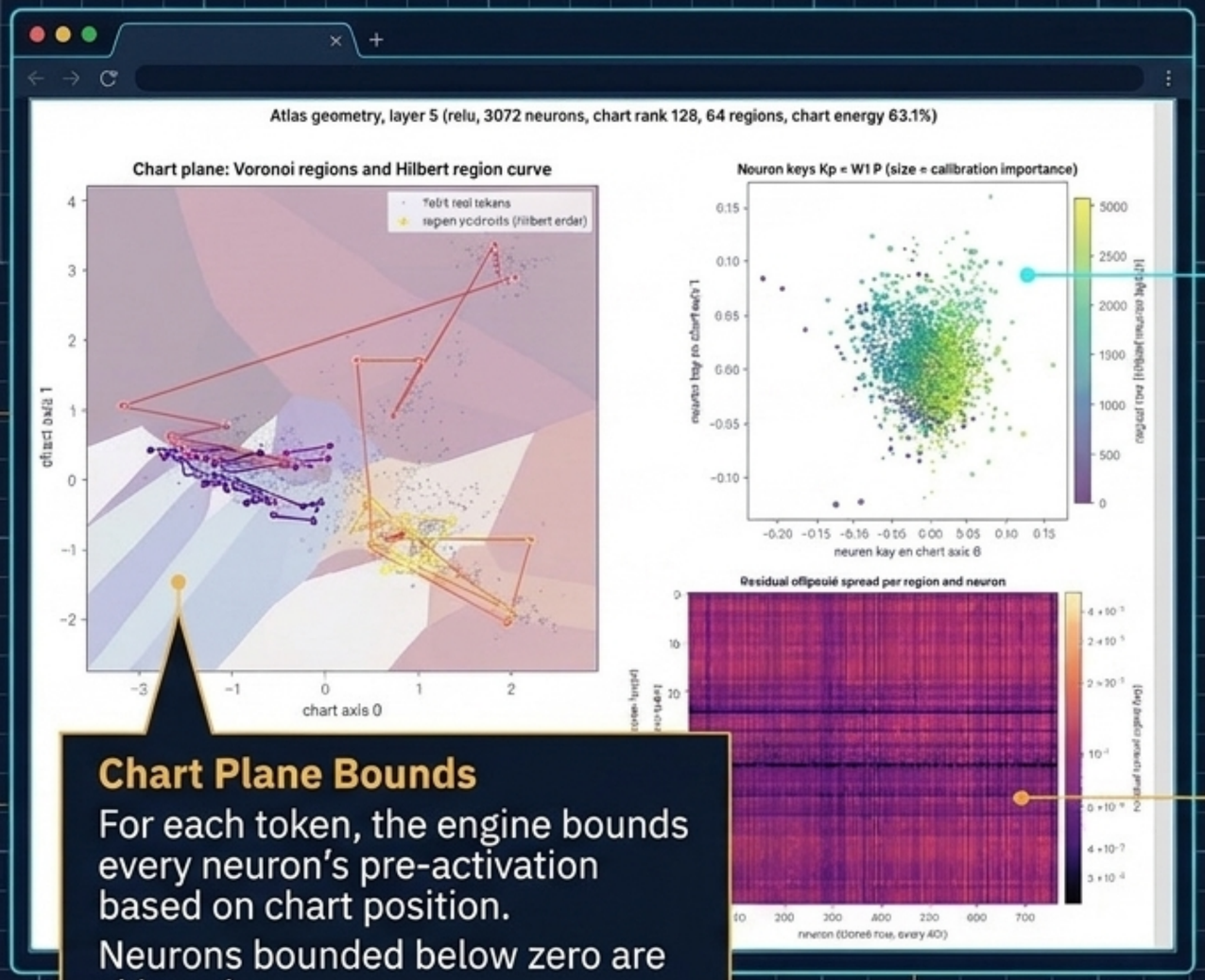


Measured on OPT-125m (32-40 tokens, prompt: 'The capital of the Netherlands is')			
Backend: torch (dense)	Decode: 24.3 tokens/s	MLP Work: 100%	Output: reference
Backend: atlas, NumPy (z 3.5)	Decode: 14.9 tokens/s	MLP Work: 37.7%	Output: identical
Backend: atlas, OpenCL (PoCL CPU, z 3.5)	Decode: 10.6 tokens/s	MLP Work: 37.7%	Output: identical

Note: The chart plane, neuron key scatter, and residual spread heatmap are static figures generated by `holoqubed_graph.py`; label them strictly as static references, not live telemetry. On this CPU VM, atlas saves multiply-adds but is slower than dense torch in wall time.

Gemini Notebook

The Geometric Idea: Activation Manifolds Predict Neuron Firing



Low-Rank Math Mapping

Mean $\boldsymbol{\mu}$ and principal axes $\mathbf{P}$ yield low-rank coordinates

$$\mathbf{Z} = (\mathbf{x} - \boldsymbol{\mu})\,\mathbf{P}$$

Neuron's chart score is

$$\mathbf{Z} \cdot \mathbf{K_p} + \text{base}.$$

Chart Plane Bounds

For each token, the engine bounds every neuron's pre-activation based on chart position.

Neurons bounded below zero are skipped.

Bounds are statistical (z standard deviations).

Residual Expansion

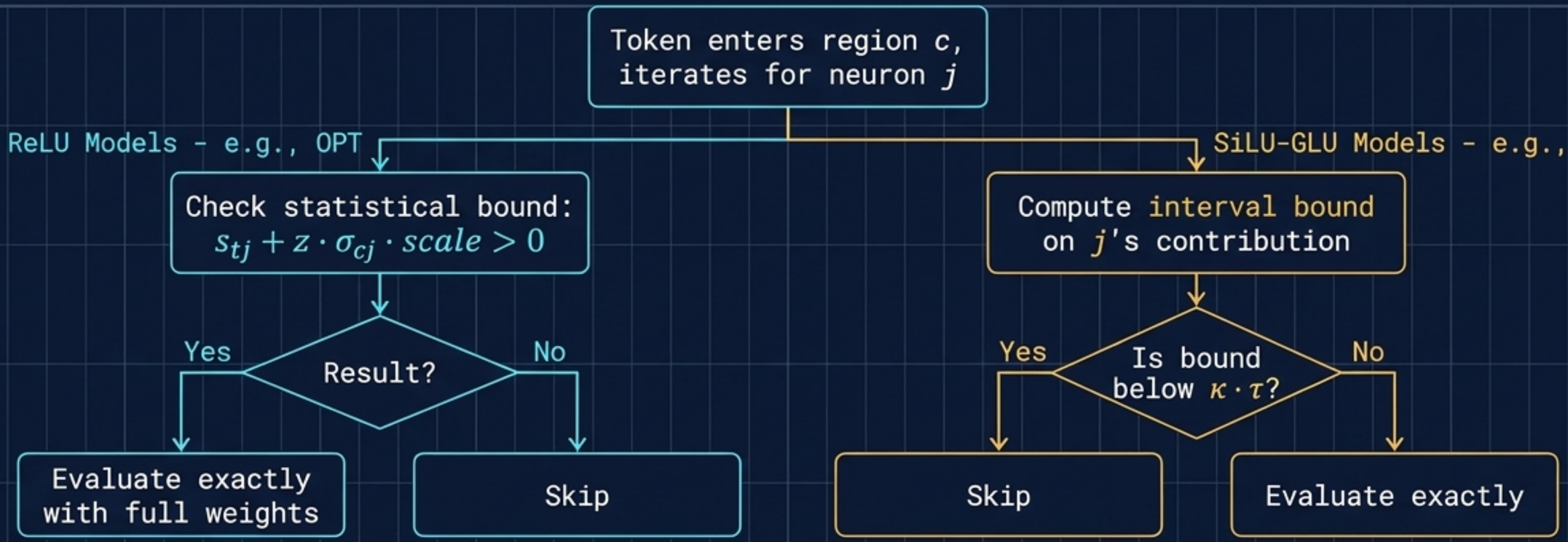
Each region stores RMS spread.

Tokens further from the typical radius widen the bound proportionally:

```
scale = max(1, |x_res| / r_typ).
```

Gemini Notebook

Routing Rules and The Hilbert Layout Advantage



The Hilbert Layout Advantage

N-dimensional Hilbert curve reordering ensures co-firing neurons sit in the same memory blocks.

At **z 3.5** on **OPT-125m**, the 16 KB blocks touched per token **drop by 17%** for W2T and **8%** for W1.

Gemini Notebook

Weights as Quantum States: Mapping Linear Maps to Statevectors

Every 4×4 weight tile is a 4-qubit real-amplitude state.

16 weights are normalized into 4 qubits using 15 RY and 14 single-control CNOTs.

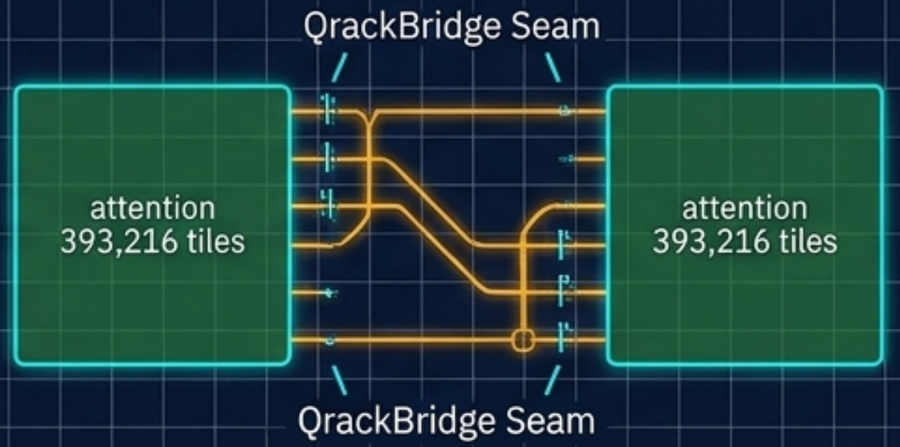
Goal: hold linear maps inside statevectors.



Seam Truncation Insight:

Seam truncation equals the ideal rank-2^b SVD (Eckart-Young) to 4 decimals.

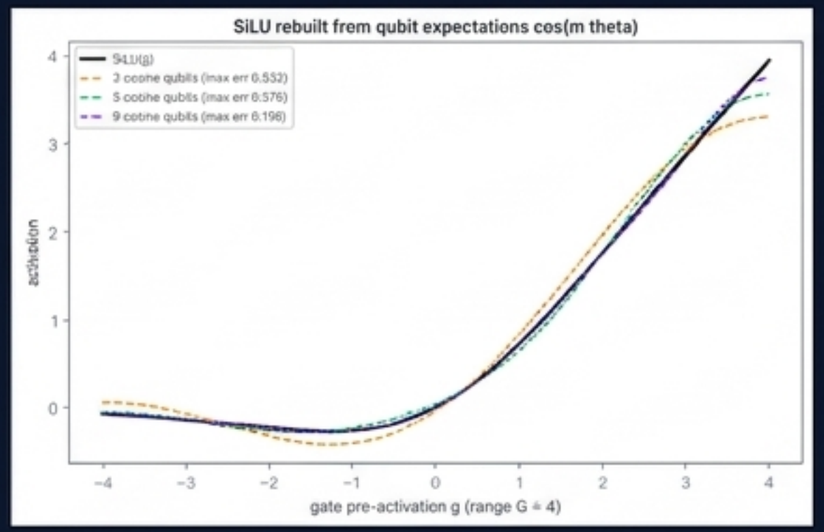
$\text{max_bonds} = K - 2$ yields 0.9999 (layer 5) and 0.9990 (layer 11) fidelity.



Gemini Notebook

qneuron Gating: Rebuilding SiLU from Qubit Expectations





SiLU is replaced with a QrackNeuron circuit: 8 gate cosine qubits + 1 constant term + 1 up qubit carrying RY($\pi/2 - \varepsilon u$).

	Cosine + Up	Mean MLP error	PPL	Top-1 vs dense	KL
	5 + 1	0.192	34.39	69.4%	0.288
default	9 + 1	0.050	27.69	89.6%	0.043
	13 + 1	0.023	28.67	89.0%	0.032

Limit Analysis: The gating mechanism sets the quality ceiling for the quantum format, peaking at ~90% top-1 agreement with dense architectures, independent of tile precision.

Gemini Notebook

Signed Readout: Recovering Sign Data via Hadamard Test



1. **The Setup:** Chart register and ancilla sit in patch 0; weight state sits in patch 1.

2. **The Query:** Controlled inverse chart preparation runs on anc=1. Anti-controlled reference flip runs on anc=0.

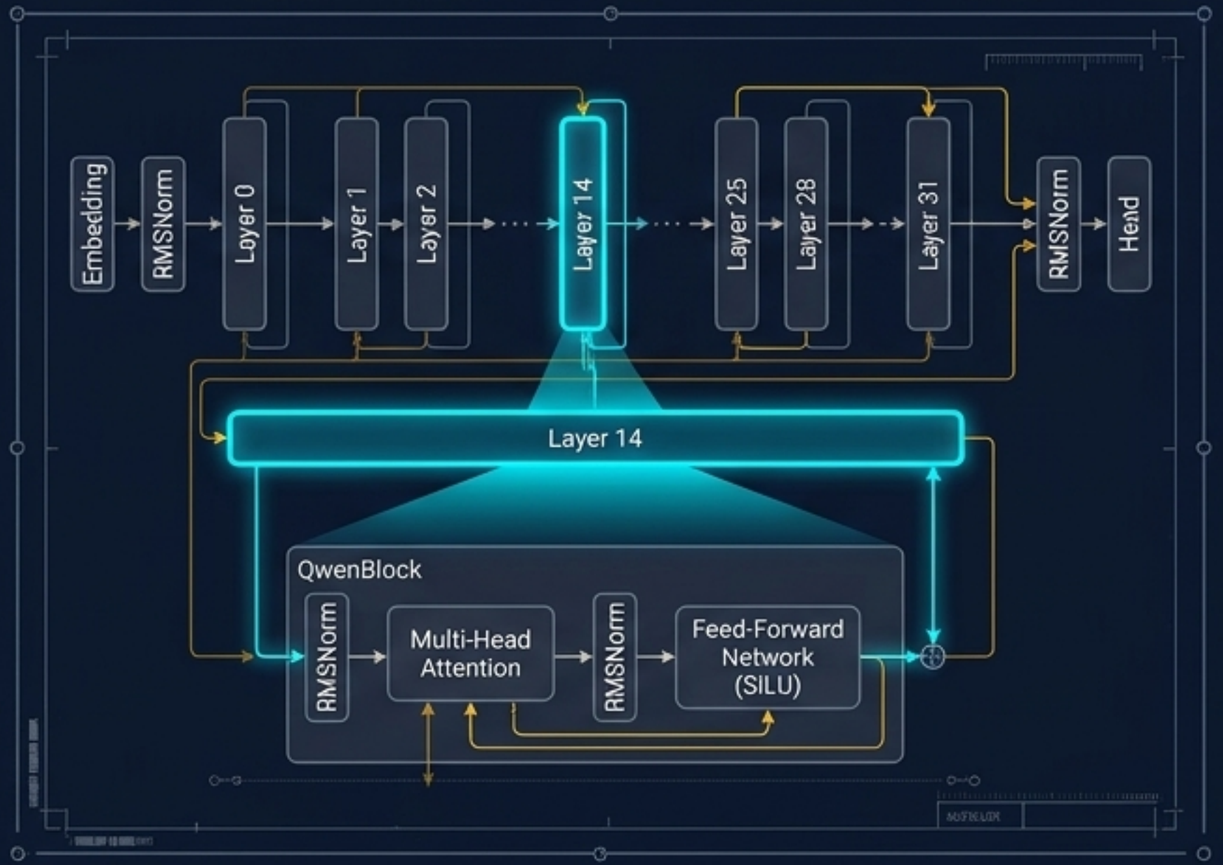
3. **The Extraction:** Post-selecting chart register on |0⟩ yields the signed score:
$$P(\text{anc}=0, j) - P(\text{anc}=1, j) = \beta \frac{s_j}{\|W\|^2 \|z\|}.$$

Readout	Positive recall	Positive precision
Exact statevector (full bonds)	99.2-100%	100%
65,536 shots	76-88%	47-74%

Conclusion: Exact readout supplies required signs for ReLU routing. Shot-based readout is not yet reliable enough for routing.

Gemini Notebook

System Emulation Report: A Genuine Layer in PyQrack



One full Qwen layer runs genuinely in PyQrack, reproducing the emulation token for token. Linear maps run as 4x4 qubit tiles; SiLU gating runs as qneurons. Loom links (RMSNorm, RoPE) remain classical.

Measured: Layer 14 PyQrack Isolation	
Tokens vs emulation:	identical
Max logit difference:	6.0e-5
Tiles in the layer:	983,040
Gates (2 tokens):	125,829,120 in 782 s
Live qubits at a time:	4 per tile, 9 per qneuron (reset after each)
Whole model (estimate):	37.2M tiles/token, ~4.7 hours/token.

Gemini Notebook

Storage Precision Analysis: Tree Angles vs. Quantization

Size: 440M layer weights

Scheme	Size	PPL	Top-1	KL
1. Baselines				
bf16 source	881 MB	28.02	100%	0.0000
fp16 tiles	881 MB	28.03	100%	0.0000
2. Int Quantization				
int8 weights	447 MB	28.06	96.47%	0.0037
int6 weights	385 MB	28.37	92.35%	0.0125
3. Angle Formats				
tree angles	420 MB	28.18	95.88%	0.0054
Gray-code	261 MB	3002	14.90%	5.1142

- Gray-code circuit angles quantize poorly (leaf angles span ~10 rad; root error moves 16 weights).
- Tree angles spread error evenly, recovering the loss efficiently.
- Tile form trails block-scaled quantization by ~0.5 bit/weight (sound for circuits, not compression).

Gemini Notebook

Geometric routing pays off on ReLU models, not yet on SwiGLU.

OPT-125m & 350m (ReLU)

Setting: z 3.5

MLP work (125m): 36.6%

Neurons routed: 46.2%

PPL (125m): 48.55 vs 48.37 (dense)
PPL (350m): 36.04 vs 35.81 (dense)

Top-1 vs dense: 99.12%

Static 50% pruning PPL: 76.94

Qwen3-0.6B (SiLU-GLU)

Setting: κ 1.0

MLP work: 108.9%

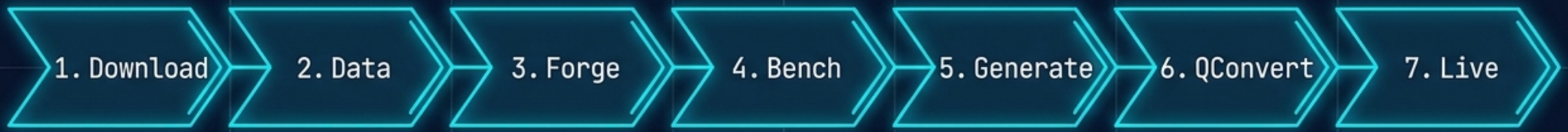
Neurons routed: 99.2%

PPL: 28.03 vs 28.02 (dense)
Top-1 vs dense: 99.61%

Analysis: SiLU activations are dense. The interval bound only becomes selective once output quality drops. Routing savings on SwiGLU require an explicit ReLU-fied gate.

Gemini Notebook

Deployment Pipeline & API Interfaces



A fresh pipeline run reaches the dashboard in 30m 25s (local hardware).

holoqubed_atlas.py serve exposes OpenAI-compatible endpoints with SSE streaming.
Backends: CPU and OpenCL.

```
POST /v1/chat/completions
    stream=true,
    extra_body={"atlas_z": 3.5, "atlas_kappa": 1.0}
```

Responses return usage.atlas reporting routed neurons and MLP work.

Gemini Notebook

Master Schematic: Classical Savings & Quantum Precision

High-Fidelity Work Reduction

OPT-125m MLP work drops to 36.6% (z 3.5) while maintaining 99.1% top-1 agreement.

(Dense)	(Atlas)
100% work | 100% top-1 | 48.37 PPL	36.6% work | 99.1% top-1 | 48.55 PPL

Dynamic routing outperforms Static 50% pruning (76.94 PPL).

Stacked Validation Metrics

Tree angle storage: 7.63 bits/weight, PPL 28.18, KL 0.0054 (far superior to Gray-code; trails int8 slightly).

PyQrack check: 4x4 qubit tiles for 1 Qwen layer yields token-identical output, max logit difference 6.0e-5.

fp16 states: fp16 tile storage & emulated statevector both keep KL at 0.0000 (4 decimals).

Gemini Notebook

Classical savings today, an exact testbed for quantum weight formats.

- The geometric atlas currently cuts MLP compute on ReLU models to ~33% while preserving near-dense quality.
- The PyQrack integration provides an exact testbed: one full transformer layer runs genuinely in PyQrack, matching classical emulation token for token.
- Current performance: Whole model emulation takes ~4.7 hours per token on 2 vCPU. qneuron gating holds top-1 agreement near 90%.

Next Steps

1. Development of better qneuron gating mechanisms.
2. Implementation of SwiGLU routing optimizations.

Every figure above was measured on a 2 vCPU VM (OPT-125m, OPT-350m, Qwen3-0.6B, WikiText-2).

Gemini Notebook